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A Pathway to Carbon-Neutral Healthcare: Exploring the Potential of Rooftop Solar Net Metering and Carbon Credits for Rural Medical Facilities in Bangladesh

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ABSTRACT

The study conducts a feasibility study on implementing a rooftop solar net metering system (NMS) it the rural healthcare facilities in Bangladesh. The study also addresses shortfalls in the existing Net Metering Guidelines of Bangladesh and suggests several improvements based on the findings that could promote the adoption of net metering across the country. The non-urban areas of Bangladesh face significant challenges in the reliable and consistent supply of electricity. To mitigate the power deficit, the integration of renewable energy sources with healthcare infrastructures could play a vital role in ensuring a seamless power supply while reducing the carbon footprint. A cost-benefit analysis has been performed to assess the viability of the project that evaluates the net present value (NPV), payback period, levelised cost of electricity (LCOE) and internal rate of return (IRR). However, the most pivotal aspect of this model is the integration of revenue from carbon credit sales, which could be a game-changer in promoting renewable energy transition for Bangladesh. The integration of revenue from environmental attributes (carbon credits) constitutes a novel financial model that has not been previously applied to the rooftop solar power sector in Bangladesh. The study also considers the potential reduction of CO_2 emissions, which could have a significant positive impact on the environment. The approach offers cost-effective, eco-friendly and sustainable energy solutions for the developing countries like Bangladesh.

1 | Introduction

Bangladesh is a densely populated country in the world and is home to approximately 171.46 million people, with the majority of them residing in rural areas [1]. Therefore, rural healthcare in Bangladesh stands as the cornerstone of the nation's overall well-being. In comparison to the rural population, the number of healthcare facilities is notably insufficient. Currently, there are only 5424 rural healthcare facilities in Bangladesh [2] serving 60% of the country's total population [1]. Moreover, the rural healthcare sector grapples with significant challenges, such as lack of infrastructure, scarcity of skilled workforce, inadequate financial resource allocation, etc. [3]. In addition to these, ensuring a reliable supply of electricity presents another pressing concern in rural areas of Bangladesh [4–6]. Rural healthcare in Bangladesh heavily relies on conventional electricity sources, leading to a significant issue of frequent power cuts due to an overloaded electricity network [5], particularly during summer. However, despite significant progress being made in scaling up its generation capacity [7], the rural distribution infrastructure has still lagged behind [5].

Ensuring an uninterrupted and quality supply of electricity is crucial for these healthcare infrastructures to maintain reliable

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TABLE 1 | Future power generation plans [Source: Iepmp 2023] [21].

Year	Generation capacity (MW)	% of clean energy	% of RE	RE installed capacity (MW)
2030	40,000	18%	5.70%	2280
2041	60,000	40%	8.80%	5280

operation of clinical equipment, support remote video consultations and deliver essential health services. Moreover, these facilities are geographically distributed across rural areas with vulnerable grids, and improving their energy reliability aligns with government goals to reduce power sector costs while increasing investment in public health.

Power generation of Bangladesh heavily relies on the fossil fuels [7, 8] like natural gas, coal, HFO and HSD, with hydroelectric power making only a marginal contribution. The depletion of gas reserves [9], the unstable LNG market with escalating costs [5], the dollar crisis, sub-standard grid infrastructure and the inadequate integration of renewable power sources are the root causes of energy insecurity in Bangladesh [9–12]. Therefore, to develop sustainable, reliable and clean power and energy systems, there is a growing emphasis on transitioning to renewable energy sources to diversify the energy mix and reduce dependence on non-renewable resources [13]. As of May 2025, renewable energy accounts for only 4.98% of the country's total installed generation capacity [14]. However, in rural areas, off-grid solar home systems (SHS) and small scale stand-alone solar power generation systems play a vital role in providing electricity to the households [15]. Implementing distributed generation systems at a mass scale, such as grid-connected PV systems, could effectively address the increasing electricity needs and help overcome the persistent crisis of energy insecurity [16, 17].

Bangladesh's low-lying geography and dense population make it particularly vulnerable to rising sea levels, flooding, and extreme weather events, exacerbating socio-economic challenges. Bangladesh ratified the Paris Agreement during COP21 in 2015, and committed to reduce greenhouse gas (GHG) emissions with an unconditional mitigation target (business-as-usual) of 5% (27.56 $MtCO_2e$) and a conditional target of 15.12% (61.9 $MtCO_2e$) by 2030 [18–20]. The updated NDC 2021 pledges to cut emissions by 18.94 $MtCO_2e$ from the energy sector, 3.42 $MtCO_2e$ from the transport sector, and 5.19 $MtCO_2e$ from the industry sector by 2030 [19]. So far, the government of Bangladesh has formulated several policies and plans, such as the Renewable Energy Policy (REP) 2008 and the Power System Master Plan (PSMP) 2016, promoting renewable power generation. Despite some progress being made, the target was not fully achieved. However, in alignment with NDC, the government has formulated a latest master plan referred to as integrated energy and power master plan (IEPMP) 2023, with ambitious targets to generate 18% (5.7% from renewable) and 40% (8.8% from renewable) of its electricity from 'clean energy' sources by 2030 and 2041, respectively [21]. Table 1 outlines the projected power generation plans for Bangladesh as per the IEPMP 2023.

The geographical location of Bangladesh offers solar energy as an alternative power source which holds immense potential to yield both economic and environmental benefits [10, 22, 23]. Since solar is a limitless and clean form of energy, it offers a sustainable solution to the growing energy consumption [24]. However, due to the country's heavy reliance on agriculture, the scarcity of land poses a significant challenge for implementing solar-powered projects. To overcome this challenge, rooftop solar grid-tied systems with net metering could emerge as a viable solution. These systems not only enable solar panel owners to reduce electricity expenses but also allow surplus power to be supplied to the grid, enabling earning credits [24]. The advantages of rooftop solar generation are manifold: (a) it addresses peak power and energy deficits, (b) provides household power during grid failures, (c) enhances energy efficiency by eliminating transmission and distribution losses and (d) enables utilities to purchase power at lower rates compared to the short-term electricity market. Furthermore, such initiatives contribute to reducing fossil fuel reliance, lower CO_2 emissions and enhance energy security through decentralised power production [25–27]. To maximise the benefits of rooftop solar systems, clear policies, grid improvements, public awareness campaigns and financial incentives are imperative for making solar panel installations and net metering more accessible, thereby fostering a sustainable and resilient energy future in Bangladesh. However, in order to foster the growth of solar power generation, the government of Bangladesh has taken several initiatives, such as the reduction of import duties and VAT exemptions on solar panels, inverters and related equipment, which make solar installation systems more affordable and accessible across the country. Moreover, the government has recently introduced the Net Metering Guideline-2025, updating the 2018 version to allow three-phase consumers, remove the 10 MW project cap, and permit installations up to 100% of the sanctioned load. Nevertheless, the latest policy does not fully address key financial and operational issues, including consumer control, billing disputes, excess power marketing limits, etc. There are few studies on solar net metering systems (NMS) in Bangladesh, focusing mostly on a single building or facility. However, none have comprehensively analysed the collective techno-economic impact on the rural healthcare sector, nor have they evaluated the financial viability of carbon credit sales as an integral part of the cost-benefit analysis for such a project. This study focuses on the utilisation of solar-based NMS, especially for the rural healthcare systems in Bangladesh, which is aligned with its Paris Agreement NDC targets, ensures uninterrupted power supply through clean energy, utilises limited land resources efficiently and leverages carbon credit opportunities. Incorporating revenue from environmental attributes (i.e., carbon credit) introduces a unique financial model, representing an approach not previously implemented in the rooftop solar power sector of Bangladesh. The primary objective of this study is to conduct a techno-economic feasibility assessment of rooftop solar NMS for rural healthcare facilities in Bangladesh, and the secondary objective is to model the financial impact of integrating carbon credit revenue to maximise the benefits of green energy, and thirdly, to propose specific, actionable amendments to the 2025 Net Metering Guidelines based on the study's techno-economic and policy findings. The key contributions of this study are summarised below:

- The study performs the techno-economic assessment of rooftop solar NMS for rural healthcare facilities in Bangladesh, incorporating carbon credits.

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- Evaluates the financial impact of integrating carbon credit revenue to maximise benefits from green energy.
- Promotes a clean, sustainable and low-carbon energy transition in Bangladesh.
- Encourages large-scale adoption of renewable NMS power generation across Bangladesh.
- Alleviates the burden on the national grid while reducing reliance on fossil-fuel-based generation.
- Offers utilities and investors with evidence-based guidance for scaling up renewable NMS initiatives.
- Reviews the 2018 and 2025 Net Metering Guidelines to identify gaps and to recommend improvements for policy enhancement.

2 | Background

Over the last few decades, Bangladesh has seen a significant rise in energy consumption due to several factors, including urbanisation, economic development and population expansion. From 1990 to 2022, the power sector of Bangladesh was predominantly dependent on fossil fuel-based energy sources. From 2000 to 2022, power generation from natural gas saw a remarkable increase in its share from 3.6% to 83.7% [28]. Since 1990, a single hydro-electric power plant had been the sole contributor to the renewable energy landscape [29], until 2015, when the government started integrating solar energy into the power generation mix. Since then, solar energy's share has notably increased in the renewable energy sector.

In 2022, CO_2 emissions in Bangladesh increased by nearly five times compared to levels from 2000, rising from 20 million metric tonnes to 104 million metric tonnes [30]. Notably, the industrial sector witnessed a substantial surge of 490%, while residential emissions grew by 207% during this timeframe. Bangladesh is expected to be one of the top emitters of CO_2 from fossil fuel power plants in the fiscal year 2023–2024 [31]. In addition to this, Bangladesh has seen an average of 8% annual growth in CO_2 emissions over the past 2 (two) decades on a global scale [20]. The significant growth of the apparel industry, which heavily relies on energy produced from fossil fuels, consumes approximately 26% of the total energy supply [32]. In FY 2021–22, this industry consumed 367 billion cubic feet (bcf) of gas, which accounts for 36% of the national gas consumption [33].

As of May 2025, the installed generation capacity of Bangladesh is 31,258 MW (including captive generation) [14], which is still dominated by fossil fuels with 39.6% from gas, 23.0% from coal, 18.6% from heavy fuel oil (HFO), 9.0% from captive sources, 3.7% from imported sources, 0.9% from high-speed diesel (HSD) and a little contribution of 4.98% from renewable energy [14].

2.1 | Current Status and Prospect of Renewable Energy of Bangladesh

Bangladesh has the potential for several renewable resources, including solar, hydro, wind and tidal energy. However, due to its geographical location (between 20° 34' and 26° 38' north latitude), solar power is the most promising source of renewable energy [11, 26, 34], with solar irradiation ranging from 4 to 7 kWh/m^2/day; while tidal and wind energy exhibit potential, especially in the coastal areas. Despite its huge prospects, the country has yet to fully harness these abundant resources. Figure 1 below shows the electricity generation fuel-mix of Bangladesh (including captive power generation) as of May 2025.

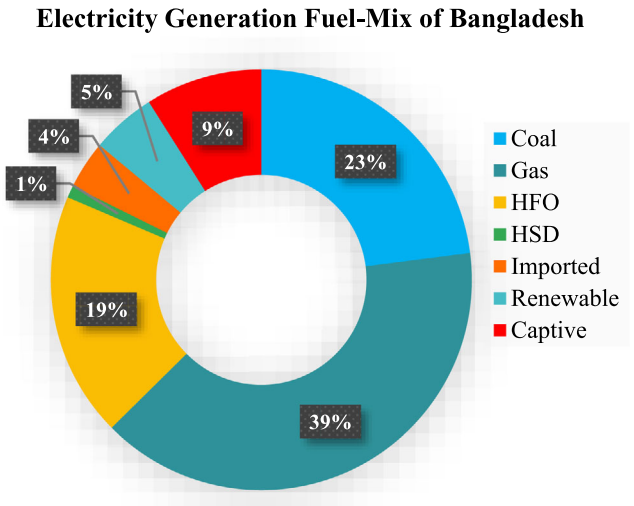


FIGURE 1 | Installed generation capacity fuel-mix of Bangladesh [Source: SREDA, accessed on May 2025].

Bangladesh has an average theoretical solar potential of approximately 4.59 GHI (global horizontal irradiance) as the country ranks in the middle range globally [35] in terms of solar radiation. Among the regions of the country, Chittagong division has the highest solar irradiation (more than 1800 kWh/m2 /year) whereas Khulna, Barishal and Rajshahi divisions have moderate solar irradiation. However, the Rangpur division has the lowest solar irradiation, with around 1500 kWh/m2/year. Therefore, the maximum to minimum solar irradiation variation is less than 20%. The solar irradiance (kWh/m2) and long-term average of GHI from 1999 to 2018 across Bangladesh are shown below in Figure 2.

As of May 2025, the installed capacity of grid-tied renewable energy is 1179.2 MW [14] and off-grid is 380.24 MW [14]. The entire renewable energy portfolio of Bangladesh is mainly dominated by solar energy. Currently, there are 14 solar, 1 60 MW wind and 1 230 MW hydro-electric grid-tied power plants operating across the country [14, 36]. The off-grid infrastructure comprises SHS, solar micro-grid, solar mini-grid, solar street light, solar powered telecom BTS and solar irrigation systems, mostly installed in the rural areas of Bangladesh [14]. Infrastructure Development Company Limited (IDCOL) [18, 37] is actively promoting the rural power sector of Bangladesh through off-grid power systems. As of 2019, IDCOL has installed 4.13 million SHS, 26 solar mini-grids, 1130 solar irrigation pumps, energy-efficient improved cook stoves (ICS), etc. across the rural areas [38], playing a vital role in fostering socio-economic development of non-urban areas of the country. Table 2 below shows the share of electricity production from different renewable energy sources:

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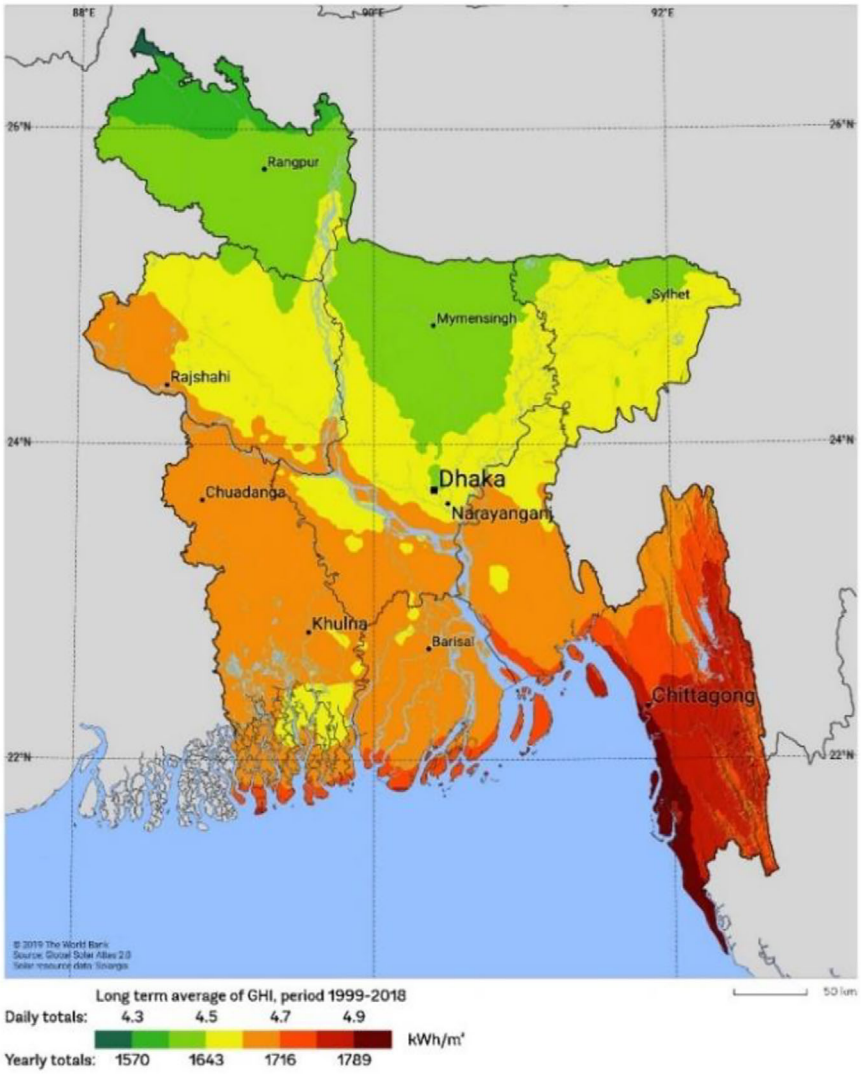


FIGURE 2 | Solar irradiation map of Bangladesh.

TABLE 2 | Renewable energy generation sources [Source: SREDA, accessed on May 2025] [14].

Energy source	Off-grid (MW)	On-grid (MW)	Total (MW)
Solar	377.15	888.12	1265.27
Wind	2	60.9	62.9
Hydro	0	230	230
Biogas	0.69	0	0.69
Biomass	0.4	0	0.4
Total	380.24	1179.02	1559.26

2.2 | Potential of Rooftop Solar Energy in Bangladesh

Rooftop solar energy in Bangladesh offers a promising pathway for decentralised and sustainable electricity generation [39, 40]. By tapping into this potential, Bangladesh can bolster energy security, reduce dependence on conventional power sources, mitigate GHG emissions, and foster community empowerment. Rooftops of public buildings, government offices, schools, hospitals and universities offer a huge scope for producing energy from solar-powered net-metering systems [41, 42]. The industrial sector, particularly commercial buildings of garment factories and industries in EPZs also present opportunities to install rooftop solar power systems. Rooftop solar generation offers several key advantages such as; addressing peak power and energy deficits, providing household power during grid failures, eliminates transmission and distribution losses, reducing carbon footprint and enabling utilities to purchase power at lower rates, benefiting both consumers and the grid [39–42].

2.3 | Carbon Emission by Energy and Power Sector of Bangladesh

As the need for electricity in Bangladesh grows, the rate of GHGs emissions is also rising. For every 132,520 GWhr of power generated, approximately 83,000 million metric tonnes (Mt) of CO_2 are emitted [31]. The latest report from the Joint Research Center (JRC), EU suggests that in 2023, Bangladesh was

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responsible for emitting 281 Mt of C02eq, which was 0.32% of the world's total emissions [43]. It also indicates that emissions from the power industry have experienced a whopping surge of 300% in 2023 compared to 1990. Moreover, according to the IEA report, in 2022 the CO_2 emissions from fuel combustion in Bangladesh stood at 104 Mt in total, and the power sector alone emitted 60 Mt of CO_2 which is 58.1% of the country's total emissions [30]. Out of 60 Mt CO_2 emissions, fuel combustion comprised 53 Mt from natural gas, 40 Mt from oil and 11 Mt from coal [30]. The IEA report also estimated that, the CO_2 emissions from fuel combustion rose by 420% in 2022 compared to 2000 [30]. Notably, electricity generation accounted for the largest share among all energy sources. It is estimated that if current trend of fossil-fuel based power generation continues, Bangladesh will become the 12th highest emitting country in the world with an emission of 279 $MtCO_2e$ by 2041 [44]. To address the country's growing electricity demands while mitigating CO_2 emissions, the government must utilise renewable energy sources such as wind and solar. If the government keeps its promise under the Paris Agreement, the power sector's CO_2 generation can be reduced by 43.74 $MtCO_2$ by 2030 [19].

2.4 | Certified Emission Reduction (CER) and Clean Development Mechanism (CDM) for Carbon Credit Systems

The reduction, avoidance or removal of each tonne of CO_2 emission from the atmosphere is referred to as 'carbon credits' [45]. These credits play a crucial role in mitigating climate change by allowing companies and individuals to compensate for their GHG emissions. Carbon credits are issued as Certified Emission Reduction (CER), each equivalent to one tonne of CO_2 [45, 46] under the clean development mechanism (CDM), which can be traded in both compliance and voluntary markets. CDM is a Kyoto Protocol-based framework [47] that enables emission reduction projects in developing countries to earn CER credits. These credits are utilised by developing nations to fulfil part of their emission reduction commitments outlined in the Kyoto Protocol [45, 47]. CDM generates CERs through investments in emissions-reduction projects and serves as the primary revenue source for the UNFCCC Adaptation Fund, which finances projects and programmes in climate-vulnerable developing countries, aiding in their adaptation efforts [45, 48]. While the CDM focuses on voluntary projects in developing countries, compliance markets ensure that industrialised nations adhere to emission reduction targets. Figure 3 below shows the CDM registration process.

2.5 | Compliance Market Versus Voluntary Market

The emitted CO_2 can be sold through voluntary or compliance markets in the form of carbon credits [49, 50]. The compliance markets are regulated by the framework of governments and international organisations such as the Kyoto Protocol or the EU Emissions Trading System, whereas voluntary markets are driven by the companies, corporates or individuals who choose to offset their carbon footprint and can trade with no legal obligations. According to the compliance market, the governments set limits

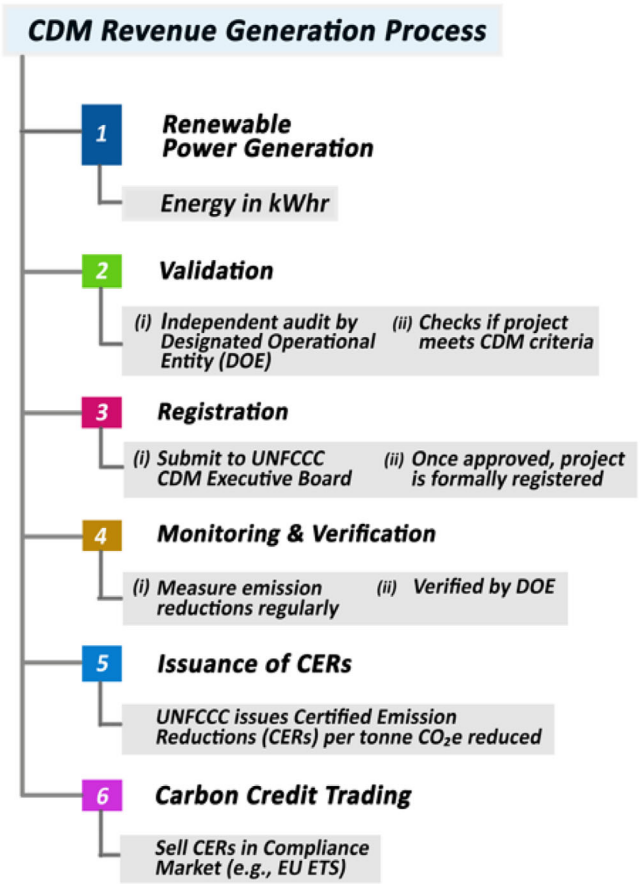


FIGURE 3 | CDM registration process.

on emissions for specific sectors, like oil, transportation, energy or waste management. Entities or agencies such as power utilities and industries falling under these regulations are allocated with emission reduction obligations [51, 52]. In the compliance market, the standards for issuing and trading carbon credits include CDM, EU ETS, and California Cap-and-Trade [51, 53].

Voluntary carbon markets (VCM), on the other hand, are non-compliance markets; that facilitate the trading of carbon credits for individuals, companies or governments who willingly or due to corporate social responsibility (CSR) initiatives choose to offset their carbon footprints to achieve net-zero emissions targets [52, 53]. Voluntary carbon credits are not region-specific and can be accessed by any sector of the economy [52–54]. The prices of carbon credits are lower, and it is much smaller than the compliance markets. In voluntary markets carbon credits are traded as Verified Carbon Units (VCUs), and the standards for issuing and trading carbon credits include Verified Carbon Standard (VCS) (supported by Verra), Gold Standard (supported by WWF), Climate Action Reserve, etc. [53, 54].

2.6 | IDCOL'S Journey in Carbon Credit Trading

Infrastructure Development Company Limited (IDCOL), a private-sector energy and infrastructure financing body of Bangladesh, is the first organisation to trade carbon credits from its from ICS and SHS projects [37]. These two (02) projects

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TABLE 3 | Daily current carbon credit generation status as on 23rd March 2025 [grid emission factor of Bangladesh: 484 Gco2/Kwh].

Grid type	Duration	Energy generation (KWh) from RE	C02 emission reduction ($MTCO_2e$)	Units of carbon credits
On-grid only	Daily	5300	2565.36	2565.36
On and off grid		7000	3388	3388
On-grid only	Annually	19,34,500	30,782.4	30,782.4
On and off grid		25,55,000	40,656	40,656

are registered under the CDM [18, 37, 55], which quantifies the reduction of CO_2 emissions and issues CERs based on auditing and verifications. Until 2020, IDCOL had earned $16.25 million by selling approximately 2.53 million carbon credits, reducing 65,603 tonnes of CO_2 emissions [55]. As IDCOL started trading carbon credits from 2006, the unit price of carbon credits varied from less than a US dollar to US $7 based on the market demand [55]. The organisation is working to scale up its projects with a vision to generate higher carbon revenues.

2.7 | Prospect of Carbon Credit Trading for Bangladesh

Bangladesh has significant potential to earn carbon credits through various CDM projects, such as renewable energy, energy efficiency, waste management, and afforestation. The renewable energy sector of Bangladesh holds significant potential for earning carbon credits, reflecting its commitment to sustainability. Its proactive approach to renewable energy development positions it favourably for carbon credit opportunities in the global market.

As of March 2025, there are 14 solar, 1 wind and 1 hydro-electric grid-tied power plants with an installed capacity of 1,178.07 MW [14] producing 5,300 MWh of electricity daily [36], which is equivalent to the reduction of 2,565.36 MT of CO_2 emissions and the production of 2565.36 units of carbon credits monthly. With the following trends, the facilities are estimated to produce 30,782.4 units of carbon credits annually. However, considering the additional off-grid renewable facilities with an installed capacity of 386.041 MW [14], the country has the potential to generate around 3,388 units of carbon credits per month, and around 40,656 units of carbon credits annually, as shown below in Table 3.

Moreover, the apparel sector of Bangladesh is one of the largest contributors to GHG emissions. In 2022, the combined emissions from the RMG and textile sectors accounted for 27.8% of the nation's total GHG emissions, equivalent to 6044 Gg CO_2eq [20]. The UNFCCC has adopted 'Fashion Industry Charter for Climate Action' that pledges to reduction of GHGs by 2030 and achieve net-zero emissions by 2050 [56]. The Bangladesh Garment Manufacturers and Exporters Association (BGMEA) has already signed the charter and adopted Sustainability Vision 2030, which aims to reduce 30% of carbon emissions, 30% reduction of energy usage and a 20% use of renewable energy by 2030 [57]. So, besides the power sector, the apparel sector holds significant potential for carbon credit trading. Based on IDCOL's experience, the challenges Bangladesh could face for trading carbon credits are price volatility in the carbon market and limited public awareness about carbon trading opportunities. Participating in 'net-zero emissions' could promote the brand value on the global platform and attract environmentally conscious customers. Local companies, such as Square Group, Beximco, Walton and City Group, along with MNCs like H&M, Unilever, Nestlé and BATB have shown strong interest in carbon reduction and could become active participants in the carbon credit trading market.

Bangladesh should develop a policy or framework with provisions for mandatory purchases of electricity from renewable energy for its emission heavy industries and agencies. They will be provided with any of the options: purchase the energy from the renewable energy producers directly or purchase carbon credits to meet the carbon footprint reduction obligation.

2.8 | Current Status of Carbon Credit Market and Future Trends

The carbon credit market is highly volatile in nature, as the price is influenced by multiple factors, such as demand and supply, regulatory policies, project type, country, location of the project, companies and certification standards [55, 58]. In 2020, the global carbon market value was estimated at around US $214 billion [55]. As of 2025, the average unit price of carbon credits in the voluntary market is around $4.80 per tonne of CO_2e, a 20% decrease from last year due to the increased supply in the market. However, the unit price rose to around $10–$20 or even more depending on the project type and certification [59].

Despite the volatile nature of the carbon credit market, its future appears promising as the global community increasingly prioritises climate action and emissions reduction. According to a report by Allied Market Research, the global market for carbon credits is projected to experience remarkable growth from $2 billion in 2022 to an estimated $143.5 billion by 2032, with a compound annual growth rate (CAGR) of 55.5% forecasted between 2023 and 2032 [60]. Experts suggest that, by 2030, the carbon trading market could reach between $7 billion and $35 billion and might climb to $250 billion by 2050 [59].

According to the McKinsey Sustainability Report, the demand for voluntary carbon credits is expected to soar significantly, potentially rising to 100 times its current level by 2050 and 15 times its current level by 2030 on a worldwide scale [61]. Boston Consulting Group estimates that the VCM, which reached $2

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billion in 2021, is expected to continue thriving. By 2030, it could reach between $10 billion and $40 billion [62]. In contrast, compliance carbon markets are experiencing a massive surge due to stricter emission caps. For example, currently, the average price of a carbon credit at the European Union's Emissions Trading System (EU ETS) is €76.75 per tonne [63, 64].

While the market is expanding, challenges like questionable emissions reductions and limited pricing data make it hard for buyers and suppliers to navigate the market. A transparent, verifiable and environmentally robust VCM is essential to effectively address global emissions reduction efforts.

2.9 | Solar Net Metering System

Primarily, there are three (03) billing programmes: feed-in-tariff, net metering, and net purchase & sale that enable homeowners to sell their PV electricity to the grid. A feed-in-tariff requires utilities to buy all PV power at a set price, while households purchase used power at regular rates. In net purchase and sale systems, excess PV electricity is purchased at a fixed price, with utilities only buying the difference when generation exceeds consumption.

Net metering allows users to offset their PV generation against their consumption, reducing electric bills regardless of when the generation occurs. In a solar PV net-metering system, if the solar panels generate more power than the consumer uses, the excess power is sent back to the grid and recorded. An energy meter tracks the electricity coming in from the grid and going out from the consumer's system. At the end of the fiscal year, the billing system adjusts for this incoming and outgoing power [16, 17, 22, 24, 27, 41, 42, 65]. If we assume the solar panel production is 'P_{pv}' and the consumer's demand load is 'P_{Load}' this relationship can be expressed as follows [26]:

$$\Delta P = P_{pv} - P_{Load} \quad (1)$$

From the equation it can be deduced that, if $\Delta P > 0$; then the solar power is exported to the grid, and if $\Delta P < 0$; then the power is imported from the grid. The rate at which the customer is billed is determined by considering various issues such as the customer tariff class, type of renewable energy technology, installed capacity and export restrictions. A bidirectional meter is used to accurately measure both the energy exported to and imported from the grid, enabling proper billing under the NMS. Figure 4 below presents the mechanism of grid-tied solar NMS.

2.10 | Solar Net Metering Financing Model

In Bangladesh, there are limited programmes offering incentives for renewable energy projects, although various institutions provide financial backing for such endeavours. The Infrastructure Development Company Limited (IDCOL), a non-banking financial entity, extends long-term soft loans for private sector projects focused on infrastructure, renewable energy, and energy efficiency. Bangladesh Bank's green financing initiative collaborates with private commercial banks, furnishing funds to support specific products and services within the renewable energy

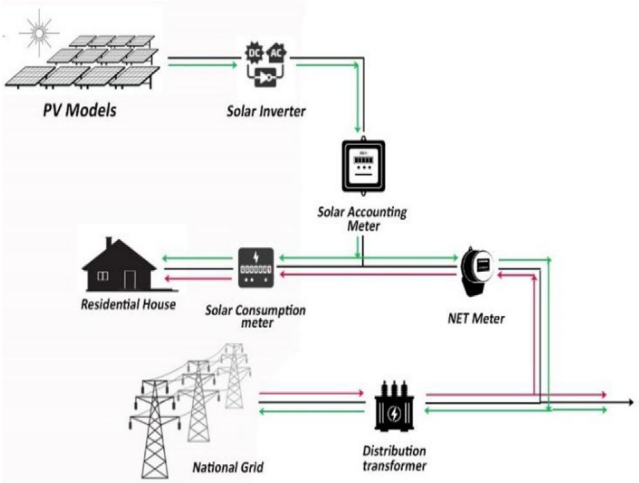


FIGURE 4 | Grid-tied solar net metering system.

domain. Furthermore, the Bangladesh Infrastructure Finance Fund Limited (BIFFL), a government-owned non-banking entity, has shown interest in financing net metering projects. Currently, two (02) financing models (i.e., CAPEX & OPEX) [65, 66] are prevalent in Bangladesh for net metering projects.

2.10.1 | CAPEX Model

Under this scheme, the power consumer assumes full financial responsibility for the investment. Typically, the consumer engages a solar engineering, procurement and construction (EPC) company, which delivers a comprehensive, ready-to-use installation of a solar power system. Following installation, ownership of the asset transfers to the consumer. Additionally, the EPC contractor may also provide maintenance services for the system under mutually agreed terms. In essence, the consumer bears all costs and owns the solar power system, while the EPC contractor manages installation and potential maintenance [66].

2.10.2 | OPEX Model

In this approach, an investor or project developer, often known as a Renewable Energy Service Company (RESCO), finances the project. The consumer then pays for the energy they utilise from the renewable energy systems installed for their benefit. Both the RESCO and the consumer enter into a long-term power purchase agreement (PPA) with agreed-upon terms and duration, specifying the tariff for the energy. In this setup, the investor owns and operates the renewable energy systems, and the consumer simply pays for the energy consumed as per the contract [66].

2.11 | Comparison Between 2018 and Latest (2025) Net Metering Guidelines of Bangladesh and Identified Gaps and Addressing Remaining Gaps

The Net Metering Guideline 2018 was the first solar rooftop energy exchange framework of Bangladesh. The earlier policy had a few restrictions, including limits on consumer type, system

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TABLE 4 | Comparison between 2018 and latest (2025) Net Metering Guidelines.

Net metering guideline features	2018 guideline	2025 guideline
Eligible consumers	Only single-phase consumers	Three-phase, prepaid, and smart meter users
Maximum system size	Up to 70% of sanctioned load	Up to 100% of sanctioned load
Maximum project capacity	Limited to 10 MW	Cap removed; allows higher-capacity projects

size and maximum capacity [66]. However, the latest version has been introduced in 2025 with few significant changes to the earlier policy [80]. Table 4 below summarises the key changes introduced in the latest (2025) Net Metering Guidelines compared to the previous (2018) version:

In the latest (2025) guidelines, several operational aspects from 2018 remain unchanged. The export of only actual power to the grid still places the reactive power burden on the grid or utilities, challenging their business model and leading to limited endorsement of net metering. Furthermore, the updated (2025) guideline does not address issues associated with the OPEX model, including limited consumer control, payment and settlement issues, restrictions on marketing excess electricity, etc. [17, 22, 23].

2.12 | Net Metering System Implementation Status in Bangladesh

According to the Sustainable and Renewable Energy Development Authority (SREDA), by May' 25, Bangladesh has installed 2,726 NMS with a total capacity of 118.021 MWp across 06 (six) power distribution utilities [14], including the Power Development Board (BPDB), Dhaka Electric Supply Company (DESCO), Dhaka Power Distribution Company (DPDC), Rural Electrification Board (REB), West Zone Power Distribution Company Ltd (WZPDC) and Northern Electricity Supply Company Ltd (NESCO). The REB has the largest installed capacity at 68.707 MWp with 438 NMS, while the BPDB has the largest number of systems installed (531), although their individual capacities are relatively smaller compared to the REB. The utilities operating in urban areas, DPDC, DESCO and NESCO have the lowest installed capacities at 3.728 MWp, 8.707 MWp and 5.62 MWp respectively, indicating a higher interest in net metering among consumers in rural areas compared to urban areas [14]. Figure 5 below demonstrates the utility-wise net-metering installation status throughout Bangladesh as on May 2025.

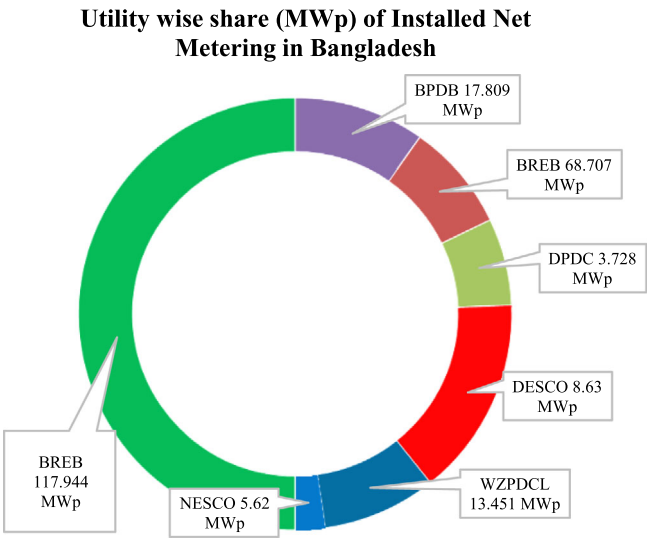


FIGURE 5 | Share of installed net metering system in Bangladesh [14] [Source: SREDA on May 2025].

3 | Literature Review

Several studies have investigated the cost-benefit and sustainability of rooftop PV systems in different regions. Arshad et al. (2015) [16] report the cost-benefit analysis on a 44 kWp rooftop PV NMS installed in Khulna, Bangladesh, which produces 53,438 kWh of electricity annually. After the consumers consume 53,010 kWh annually, the surplus amount of 428 kWh is sold to the grid at US$0.0355 per unit.

This yields monthly savings of US $403.26 and annual savings of US$2,357.11. With the 12-year capital investment recovery period, the project's overall financial benefit is US$2,760.37 over the 25-year lifetime of the solar panels. These results demonstrate the profitability and environmental sustainability of the rooftop solar net-metering system. Saquib et al. [17] conducted a study on a 25 kWp rooftop PV NMS located in Dhaka, Bangladesh, requiring a 144 m^2 area and projected to produce 611,553 kWh of electricity over 20 years, which would generate substantial savings over its lifespan, underscoring its cost-effectiveness and sustainability. With a total project cost of US$22,545.63 USD and substantial savings of US$49,415.72 USD over the same period, it offers a promising return on investment. The project's Net Present Value (NPV) was $26,881.78 USD, indicating its cost-effectiveness and sustainability as an investment. Hossen et al. [67] set up a 1 kWp grid-tied rooftop solar PV system in an academic building at Patuakhali Science & Technology University (PSTU), Bangladesh. The designed 1 kWp solar photovoltaic system has an average annual solar output of 1,236 kWh, with a total solar energy output of 30,900 kWh over its 25-year lifetime. This system contributes to a CO_2 emission reduction of 20,703 tonnes during the same period.

Alam et al. [65] performed a cost-benefit analysis on a 448 kWp net-metered rooftop solar system at Debunear Ltd., Savar, Dhaka, Bangladesh, demonstrating the financial viability of the project and its sensitivity to electricity tariffs. The system, following the CAPEX model outlined in the 'Net Metering Guidelines-

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TABLE 5 | Net-metering status in South Asia [70, 71].

Country	Eligible sector	Applicable net netering tariff
Bangladesh	Residential, commercial and industrial	Tariff rates vary based on LT Consumers [76]
India	Varies on sub- national level: all consumers are eligible majority of states except for West Bengal	Himachal Pradesh: 6–6.7 US cent/kWh. Karnataka: 9.6 US cent/kWh. Uttar Pradesh: 0.67 US cent/kWh. Punjab: same as retail supply tariff Rajasthan, Goa, and union territories as per regulated tariff
Pakistan	All residential, commercial, industrial and agricultural consumers having three- phase connection	1.2-9.4 US cent/kWh
Sri Lanka	Residential commercial and industrial	12 US cent/kWh for first years, 8.3 US cent/kWh for 8–20 years
Nepal	Domestic, organisational and commercial	Same as retail supply tariff
Maldives	Small scale domestic and business	No data available

2018', incurred initial costs of US$153,315 for solar modules and inverters, along with US$45,330.83 for local EPC services, including taxes. The project's lifespan is estimated at 20 years, with an initial investment cost capped at $547.44 per kWp. In this project, the levelised cost of electricity (LCOE) stands at 0.03125 US$/kWh for 'self-finance' and 0.0369 US$/kWh for 'without self- finance'. The IRR is 13.41% for 'self-finance' and 11.13% for 'without self-finance', while payback periods are 11.12 and 14.36 years respectively. NPV stands at US$133,833.54 for 'self-finance' and US$77,526.62 for 'without self-finance,' respectively.

Shivalkar et al. [68] explored net metering for rooftop PV systems across 24 commercial buildings by Reliance Energy Limited. The feasibility study was conducted on a 15 kWp system, aimed to replace grid power, with a shadow-free area requirement of 12 m^2 per 1 kW of capacity. Economic analysis reveals that the 15 kWp system generates 22,500 kWh annually, saving costs and reducing CO_2 emissions. With a total cost of US$10,948.90 over 25 years, it is projected to cut CO_2 emissions by 19,575 kg annually. The average cost, including initial costs, maintenance, and others, is US$15,054.74, resulting in a cost of Rs 0.05 per unit. Thakur et al. [69] proposed a 'multi meter single system is a collaborative net metering model in India, where consumers unite to create a single electricity generation facility managed by the utility. Simulations for three consumer categories showed payback periods below 8.5 years and positive NPVs, proving its financial feasibility.

In both South Asia [70] and Southeast Asia [71], the adoption of renewable energy (RE), particularly solar installation under net metering policies, has gained momentum. Tables 5 and 6 below present the net metering status in South Asian and South-East Asian countries respectively.

4 | Methodology

This study conducts a feasibility study on the potential for the grid-connected rooftop PV NMS for the Union Health and Family Welfare Centres of Bangladesh. Incorporating the non-urban healthcare facilities with renewable energy electricity production can give us a competitive edge towards the implementation of a clean and sustainable energy system. The cost-benefit analysis has been performed considering the revenue from both energy and carbon credits to assess the overall feasibility of the project. Since the healthcare infrastructures are scattered across the country; the individual rooftop areas were aggregated into a single consolidated area. Similarly, for power generation and estimating installation capacity, the entire system was treated as one unified system.

The system sizing was performed using SREDA's net-metering calculator [72], a web-based tool developed by the Sustainable and Renewable Energy Development Authority (SREDA). Based on the cost of solar investment, existing tariff rates and available area the SREDA net metering calculator performs calculations for installation capacity, energy output, revenue from energy and CO_2 reduction in accordance with the Net Metering Guidelines-2018 of Bangladesh. However, this tool does not consider the revenue from carbon credits. The study integrates the sales revenue from carbon credits alongside the revenue from energy. The revenue from carbon credit was calculated based on the prediction and analysis of future market trends, assuming an increase with several CAGR% [51–54, 59, 60].

The entire process can be summarised in the following steps as shown in Figure 6 below: (1) Collect the rooftop area, (2) Measure solar irradiance, (3) Calculate the PV panel azimuth angle, (4) Collect the sanctioned load data, (5) Collect the monthly electricity consumption data, (6). Perform system sizing based on SREDA Net Metering Guideline 2018, (7) Perform Cost-Benefit Analysis, (8) Add revenue from carbon credit sale based on % of CAGR, (9) Verify accuracy with the SREDA net metering calculator (generation output only) and (10) Perform sensitivity analysis.

4.1 | Data Collection

To perform this study, data from 50 real-life healthcare facilities from different parts of Bangladesh, comprising Union level

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TABLE 6 | Net-metering status in South-East Asia [70, 71].

Country	Eligible sector with limits	Applicable net metering tariff
Thailand	Commercial, industrial and residential sectors with up to 1MWp capacity	0.25–0.34 USD per kWh
Indonesia	Commercial, residential and industrial sectors	0.00070–0.00083 USD per kWh
Philippines	Residential, commercial, industrial and institutional sectors with a capacity limit of 100kWp	0.19 USD per kWh
Vietnam	No capacity limit on the residential sector	0.00028–0.00053 USD per kWh
Malaysia	Residential, commercial, industrial and agricultural sectors with varying capacity limits.	4.62 to 12.10 USD per kWh

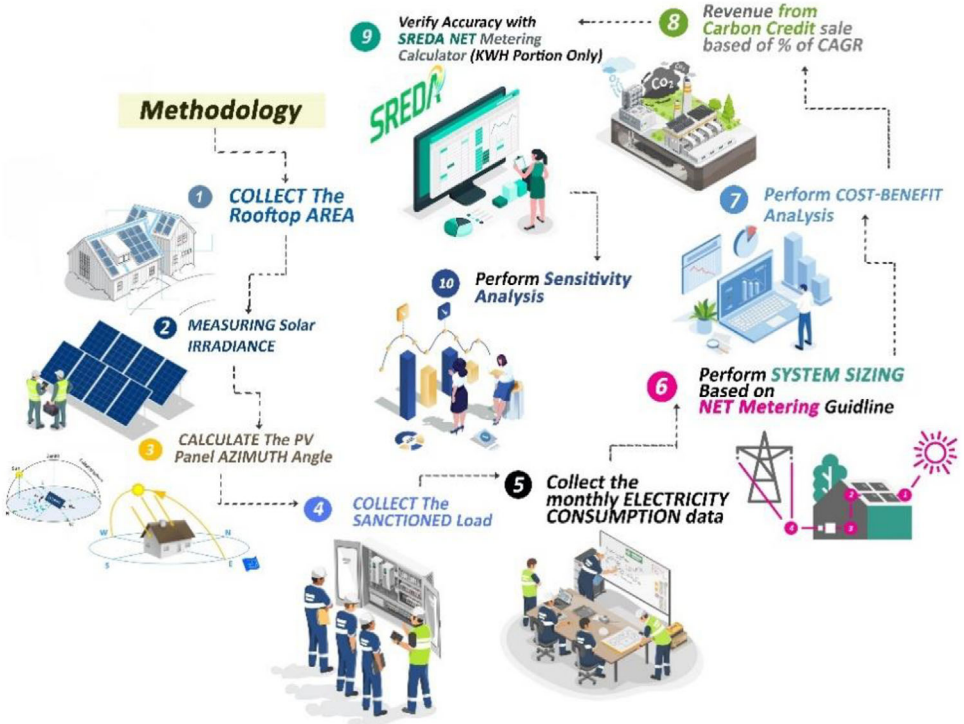


FIGURE 6 | Methodology for the cost-benefit analysis.

Health and Family Welfare Centres, 10 bed and 20 bed hospitals were collected. The sample health care facilities are Kaptai 10 bed, Kuakata 10 bed, Keraniganj 20 bed, Patgram, Santahar, Sidirganj, Rasulpur Union (Joypurhat), Keroa Union (Laxhmipur), Keaiyn Union (Munshiganj), Kotchandpur Union (Jhenaidah), Taragai Union (Mymensingh), Kamaria Union (Sherpur), Fulbari Union (Dinajpur), Sylhet, Rasulpur union (Joypurhat), Bhairab Union (Kishoreganj), Shalikha Union (Magura), etc. The collected data are comprised of: (a) size of the health care facility (in Sq-ft) for available rooftop area, (b) geographical location, (c) monthly energy consumption and (d) sanction load of the healthcare facility. Since Bangladesh is located near the Tropic of Cancer, the variation in solar irradiance from the northern to the southern regions is relatively small. Therefore, an average value of 4.59 kWh per square metre per day (kWh/m^2/day) has been considered in this study. Moreover, the rural healthcare facilities are small in size and operate during the daytime period and their energy consumption does not coincide with peak demand periods. So, there is minimal variation in load profiles across facilities. Thus, an average load profile based on the sanctioned load of each facility has been considered in this study.

4.2 | System Sizing of the Rooftop Solar NMS System

There are approximately 5424 union-level health care facilities in Bangladesh; among them, 3924 facilities are union health and family welfare centres under DGFP, 87 are union health and family welfare centres under DGHS, 1362 are union sub centres, 19 are 10-bed hospitals and 32 facilities are 20-bed hospitals. Out of 5424 facilities [2]; 5,373 facilities are usually small in size and consume less electricity. The union health and family welfare centres under DGFP and DGHS are sized between 70 and 110 sq.m. The average size of the 10-bed hospital and 20 bed hospital is around 400 sq.m and 670 sq.m, respectively. The collected

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data were collated and compiled for the analysis. These data were collected from the respective utility offices from different parts of Bangladesh. To simplify the calculation, based on the available data, the total area of all 5400 healthcare facilities was consolidated into a single unit and the estimated usable space for solar production stands at around 70,799 m^2. The solar system's installation capacity was determined based on the commonly used thumb rule of 9 m^2 per kWp [72]. Based on the available space across all healthcare facilities, the system was considered a single combined unit. With an accumulated area of 70,799 m^2 available for solar installation and a total sanctioned load of 9.30 MW, the installed capacity of the entire project is estimated at 7.86 MWp (AC). To ensure grid stability and account for weak distribution network constraints, 70% of the sanctioned load has been considered for system design.

4.3 | Methods of Calculating Solar Power

There are four methods for calculating the annual electricity generation of a solar system. The methods are: (a) average of AC output kWh/kWp/year, (b) average of AC output kWh/kWp/day, (c) plant capacity factor and (d) photovoltaic power potential (PVOUT) based on GIS data of the location. In this study, the first method (method a) (i.e. Average of AC output kWh/kWp/year method) has been adopted. Based on the solar irradiation of Bangladesh, the average output from the solar system is 1200 kWh/kWp/year [35, 72], which has been considered. However, the actual energy production from a specific solar system may vary based on its unique PV panel characteristics, local conditions and maintenance. The degradation rate of PV module efficiency was considered 99% in the first year and gradually reduced to 83% by the 20th year.

4.4 | Methods of Calculating Reduction of CO_2 Emissions

The reduction of CO_2 emissions can be calculated using the Grid Emission Factor (GEF). The GEF (gCO_2/kWh) for Bangladesh for the intermittent energy (e.g., solar, wind) is 484 gCO_2/kWh [73]. The CO_2 emission reduction was calculated by multiplying the GEF with the annual energy production (kWhr) from the solar PV system.

$$\text{Reduction of } CO_2 \text{ emission} = \text{Energy (kWhr)} * \text{GEF} \quad (2)$$

4.5 | Methods of Calculating Carbon Credit Earnings

Since the carbon credit market is volatile in nature, the study considers several CAGRs of carbon credit based on the market trends. Based on the carbon credit market trends 1.5%, 3% & 5% CAGR were considered. For the initial year, the average price of 1 unit of carbon credit was set at US$5.0, in accordance with the prevailing market rate [74]. The revenue from carbon credit sales is calculated by multiplying the amount of annual CO_2 emission reduction (in million metric tonnes) with the unit rate of carbon credit based on the prevailing market rate. The annual carbon credit revenue (US$) is calculated as:

$$\text{Reve}_{\text{Carbon Credit}}(US\$) = CO_2 \text{ Reduction (MT)} \times \text{Unit Price}_{\text{Carbon Credit}}(US\$) \quad (3)$$

The unit price for each subsequent year is adjusted using the assumed CAGR:

$$\text{Unit Price}_n = \text{Unit Price}_0 \times (1 + \text{CAGR})^n \quad (3.1)$$

Where, Unit Price_0 is the initial year price, and n is the number of years from the initial year.

4.6 | Cost-Benefit Analysis

During cost-benefit analysis, the study considered revenue from two distinct sources: (1) revenue from net energy production from the solar system and (2) revenue from the carbon credit sales based on the annual reduction of CO_2 emissions (million metric tonnes). The total cost of the project comprises of four categories: initial investment, maintenance cost, replacement cost and insurance cost. The cost-benefit analysis determined the metrics such as net profit from the project, NPV, payback period, internal rate of return (IRR) and LCOE. The average lifespan of the project was considered 20 years.

$$\text{Payback period} = \frac{\text{Total cost of the project} \times \text{year}}{\text{Annual savings}} \quad (4)$$

$$\text{NPV} = \sum_{t=0}^{N} \frac{Ct}{(1 + IRR)\,t} = 0 \quad (5)$$

$$\text{LCOE} = \frac{\text{Total cost of investment in present value}}{\text{Power (kWh) generated over lifetime}} \quad (6)$$

Where, Ct is total cash flow at time t, IRR = internal rate of return (the rate you are solving for), t = time period (usually years), n = total number of periods.

5 | Results and Discussion

In this study, a cost-benefit analysis has been carried out as part of a feasibility study to assess the potential of implementing grid-connected rooftop PV NMS for rural healthcare facilities of Bangladesh. To do so, the system sizing was estimated based on the monthly energy consumption, sanctioned load and area (sq-ft) of 50 healthcare centres across the country. Since the healthcare facilities vary in size by category (Union health and family welfare centres, 10-bed and 20-bed hospitals), the weighted average size for each category was calculated based on the number of facilities in each category. The total area of all healthcare facilities was combined into a single unit and based on the latest Net Metering Guideline-2025 (installed capacity is considered 70% of sanctioned load due to weak network infrastructure in rural areas), the potential solar installation capacity was estimated. The compiled data was fed to the SREDA net metering calculator to obtain the result. Moreover, the revenue from carbon credit sales was also considered in this study based on different growth scenarios of carbon credits as per market projection. Table 7 below presents the assumed parameters for the cost-benefit analysis, while Table 8 below illustrates the

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TABLE 7 | Assumed parameter cost-benefit analysis for the rooftop PV based NMS [May 2025] [66, 72].

Sl	Parameters	Values
1	Area to power generation factor [72]	9 m2/kWp
2	Maximum capacity of solar system as per 'Net Metering Guideline' [66]	70% of sanctioned load
3	Actual installed capacity of solar module [DC:AC ratio]	1.2 times of AC output capacity
4	Approximate installation cost per kWp system (DC capacity) (approx.)	BDT 35,000 or US$286.885
5	Component replacement cost @11th year% of the initial investment (inverter and other accessories)	25%
6	Maintenance cost (%of the initial invest)	2%
7	Insurance (percentage of initial investment per year) [72]	0.5%
8	Average of AC output per kWp solar system per year [355 Days]	1200 kWh/kWp/year
9	Retail electricity tariff rate [tariff category: LT-D1 (hospital)] [As per BERC] [76]	BDT7.55 + 5% VAT (US$0.061885 + 5.0% VAT)
10	Value added tax (VAT) on retail electricity tariff rate	5%
11	Net metering tariff for energy sales/33 kV bulk tariff for BREB] [As per Net Metering Guidelines 2018]	4.3679 BDT/kWh or 0.0358 US$/KWh
13	Grid emission factor, gCO_2/kWh of Bangladesh for solar, wind [73]	484
14	Avg. inflation/discount factor US$ [75]	2.50%
15	US$ to BDT conversion rate [April'25] [77]	122.00 BDT
16	Degradation rate of the PV module Efficiency (20 Years Period)	99% (first yr) to 83% (20th yr)
17	CAGR (compound annual growth rate) (%) of carbon credit	1.5%, 3.0% and 5.0%
18	Base value of carbon credit [74]	US$5.0

TABLE 8 | System sizing and results of the cost-benefit analysis (in present value).

Number of healthcare facilities	Total approx. rooftop area	Installed capacity of solar module	Sanctioned load	Cost of initial investment	Investment cost of project lifetime	Energy generation of project lifetime	C02 emission reduction of project lifetime
5424	70,799 Sq.m	7.86 MWp AC	9.30 MW	US$2.25 Million	US$3.59 Million	171.48 GWhr	82,999.02 $MTCO_2e$

system sizing, investment cost, energy generation, CO_2 emission reduction, etc. in present value for the rooftop PV-based NMS:

5.1 | Cost (Investment) Analysis

The costs associated with this project include (i) initial installation cost for setting up the solar panels, inverters and energy metres; (ii) maintenance cost, which is roughly 2% of the installation cost annually; (iii) insurance cost that is around

0.5% of the installation cost and (iv) inverter replacement cost required once on the 11th year of the project, which is 25% of the installation cost. Based on the current market scenario of Bangladesh, the initial investment cost of a 1 kWp solar system is roughly around BDT 35,000 or 286.88 US$/KWp. Thus, the initial investment required for the installation of a 7.86 MWp solar system stands at US$2.25 million. Over the period of 20 years, the net investment is expected to reach around US$3.59 mllion calculated in present value, considering the US inflation rate of 2.5%. However, the amount of energy (171.48 GWh) generated from the solar project, if connected to the grid without solar NMS, the consumers would have to pay around US$9.9 million (in present value) for electricity bills over a 20-year period as shown in Figure 7 below, based on the current tariff rate assuming a 4% increase in the tariff every 3 years. However, while the BERC tariff framework indicates a gradual increase of retail tariffs over time, the installation costs of solar PV systems in Bangladesh are following a downward trajectory due to recent government policy interventions that aim at reducing import duties and providing VAT exemptions on renewable energy equipment.

In line with the REP 2025, these measures suggest that future solar project start-up costs may become even more favourable, potentially exerting a positive influence on long-term financial outcomes.

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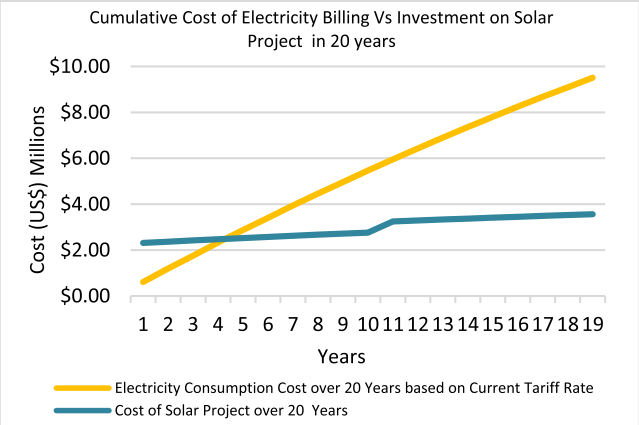


FIGURE 7 | Comparison between cumulative cost of conventional billing and cost of PV net metering system project investment in 20 years.

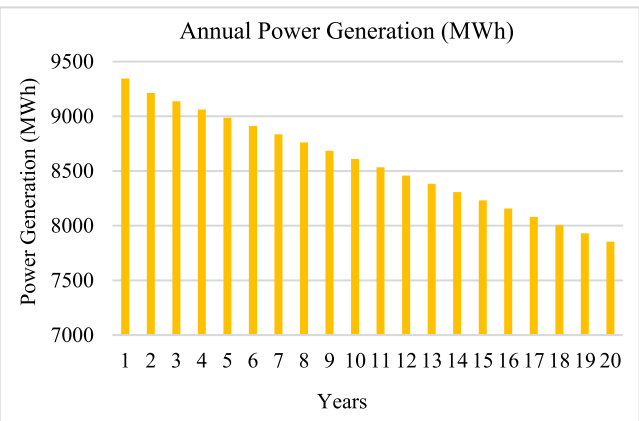


FIGURE 8 | Energy generation (MWhr) annually from solar power system in 20 years period.

5.2 | Net Energy Generation

Given the approximate rooftop area, the installed capacity of the PV-based NMS is expected to be approximately 7.86 MWp, considering utilisation of 70% of the rooftop area in accordance with the NMS Guidelines 2018. Over the span of two decades, the project is expected to generate a substantial amount of electricity, estimated at 171.48 GWhr. This estimation takes into account the average output from solar power generation, calculated at 1200 kWh/kWp/year [35], while considering the degradation rate of PV panels from 99% in the first year to 83% by the 20th year. Figure 8 below represents yearly energy generation (MWhr) from solar NMS during the lifetime (20 years).

Figure 9 below shows rate (%) of efficiency degradation of solar panels during 20 years period.

5.3 | CO_2 Emission Reduction

Based on the GEF of Bangladesh, the solar NMS makes a substantial contribution to environmental conservation by averting an estimated 82,999.02 million metric tonnes of CO_2 emissions over a span of 20 years. The GEF for intermittent energy sources

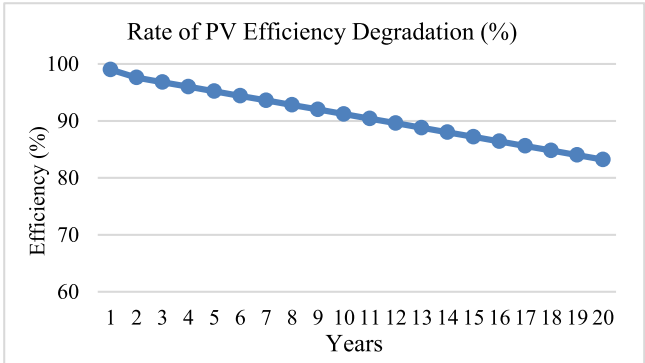


FIGURE 9 | Rate of (%) efficiency degradation of PV cell.

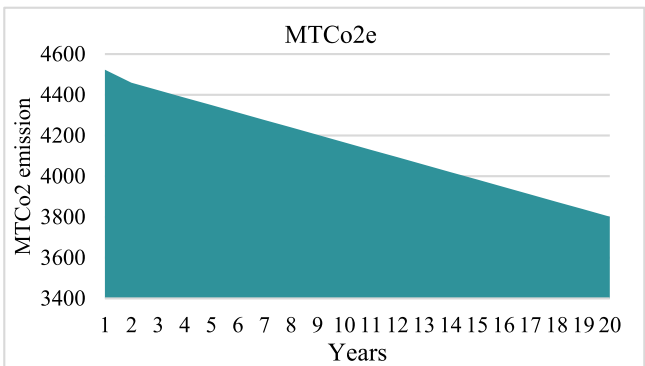


FIGURE 10 | CO_2 emission reduction (MT) annually from the project in 20 years.

such as solar and wind power in Bangladesh is 484 gCO2/kWh [72]. The CO_2 emissions, measured in million metric tonnes over a 20-year period, are illustrated in Figure 10 below.

5.4 | Financial Viability Assessment: Revenue From Solar Energy and Carbon Markets

The study considers all relevant costs, including the initial investment, maintenance cost, insurance, and cost of inverter replacement throughout the 20 years period, estimating a total present value cost of around US$3.59 million. The revenue is comprised of two components: electricity generation (Kwh) and carbon credit sales. Over the period, the system is expected to generate 171.48 GWh of energy (approx.), estimating around US$6.94 million in present value based on the current tariff rate by BREB. Since the carbon market is volatile in nature, based on future market trends the study considers several scenarios–1.5%, 3%, 5%, 0%, −10%, −25% −50% growth in carbon credit value, referred to as CAGR. Based on the growth scenarios (i.e., 1.5%, 3% and 5%), the revenue from carbon credits stands at US$0.379 million, US$0.434 million and US$0.523 million, respectively in present value. Therefore, the combined revenue (energy plus carbon credits) is projected at US$7.319 million for a CAGR of 1.5%, US$7.374 million for a CAGR of 3% and US$7.463 million for a CAGR of 1.5%, respectively, in present value terms. The LCOE also varies based on the CAGR of carbon credit price.

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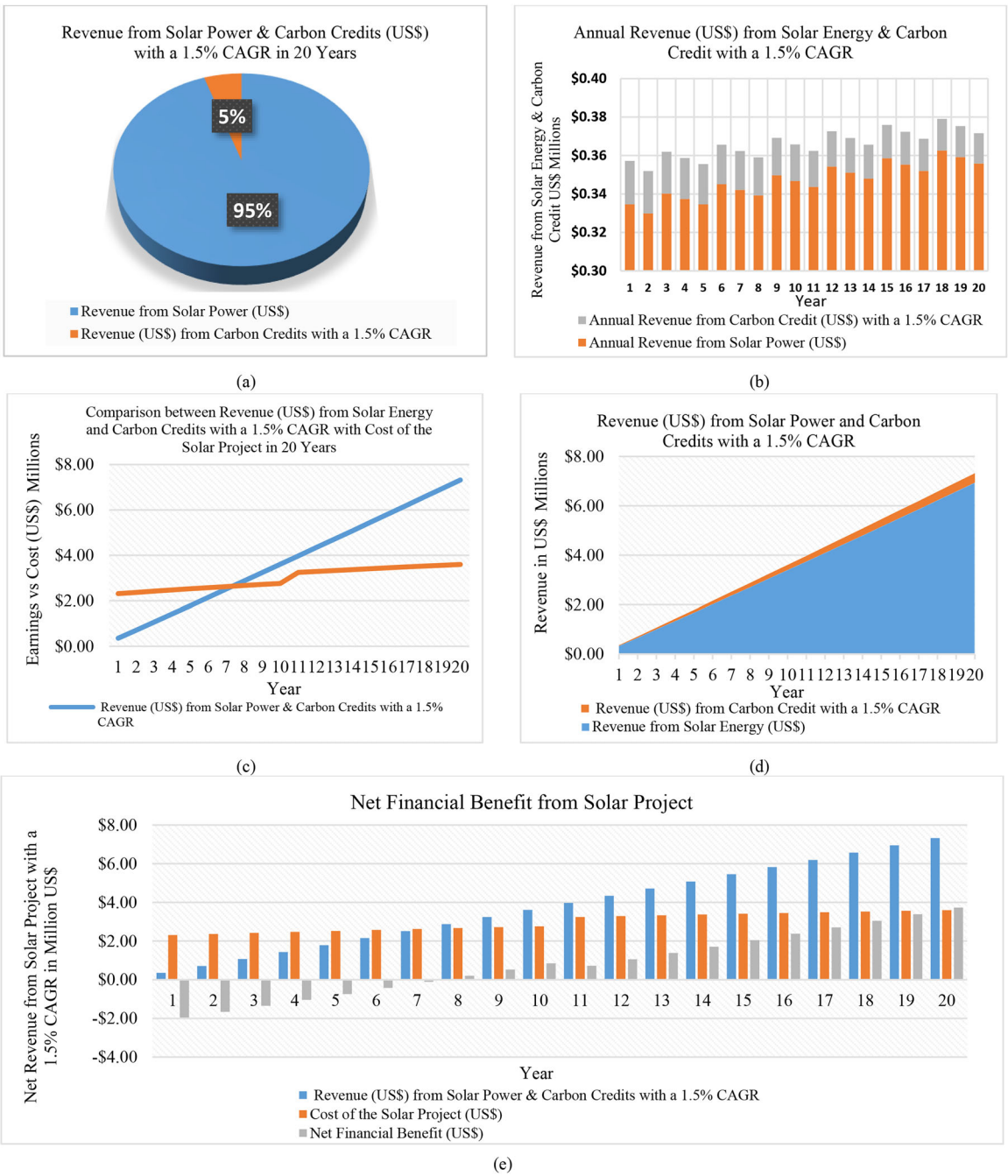


FIGURE 11 | Cost-benefit analysis with a CAGR of 1.5% of carbon credit; (a) share of revenue (%) from solar energy and carbon credit; (b) annual revenue (US$) from solar energy and carbon credit; (c) revenue (US$) from (solar energy + carbon credits) versus cost of the solar project in 20 years; (d) revenue (US$) from solar power and carbon credits in 20 years; and (e) net financial benefit (US$) with a CAGR of 1.5%.

However, due to volatility of carbon markets and regulatory uncertainties, the revenue stream from the carbon credits should be regarded as a non-guaranteed income source with inherent risk. The Figures 11–13 below illustrate the cost-benefit outcomes of carbon credits under CAGRs of 1.5%, 3% and 5%, respectively. Each figure includes (a) revenue share from solar energy and carbon credits, (b) annual revenue trends, (c) cumulative revenue versus project cost over 20 years, (d) total revenue, and (e) net financial benefit.

5.5 | NPV, IRR and Payback Period Calculation

As mentioned previously, the analysis considered revenue from solar power generation alongside earnings from carbon credits for the cost-benefit assessment. It's important to note that carbon credit prices are subject to market fluctuations, given the dynamic nature of the carbon credit market. Projections suggest a significant increase in carbon credit values over the next two decades. Therefore, the study incorporated multiple

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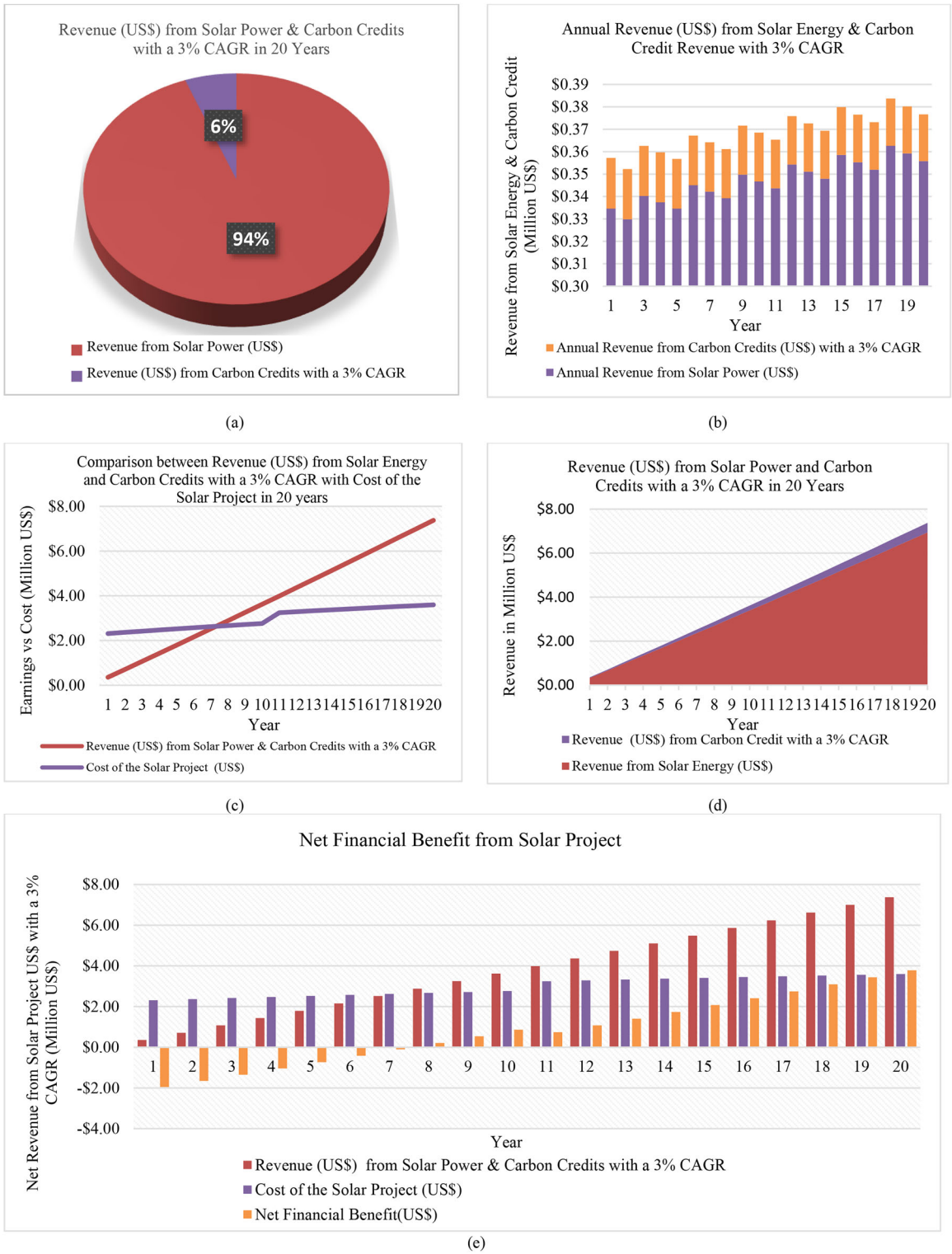


FIGURE 12 | Cost-benefit analysis with a CAGR of 3% of carbon credit; (a) share of revenue (%) from solar energy and carbon credit; (b) annual revenue (US$) from solar energy and carbon credit; (c) revenue (US$) from (solar energy + carbon credits) versus cost of the solar project in 20 years; (d) revenue (US$) from solar power and carbon credits in 20 years; and (e) net financial benefit (US$) with a CAGR of 3%.

scenarios with different CAGR% (1.5%, 3%, 5%, 0%, −10.0%, −25.0% and −50.0%) of carbon credit based on future market trends. Table 9 below shows NPV, IRR and payback period for various CAGRs of carbon credits considering 20 years of project lifetime.

It is observed that the LCOE decreases as carbon credit prices increase, whereas it rises when the CAGR or the price of carbon credit decreases. However, a substantial variation of LCOE is noticed with changing carbon credit CAGRs, reflecting the sensitivity to carbon credit market dynamics. Since

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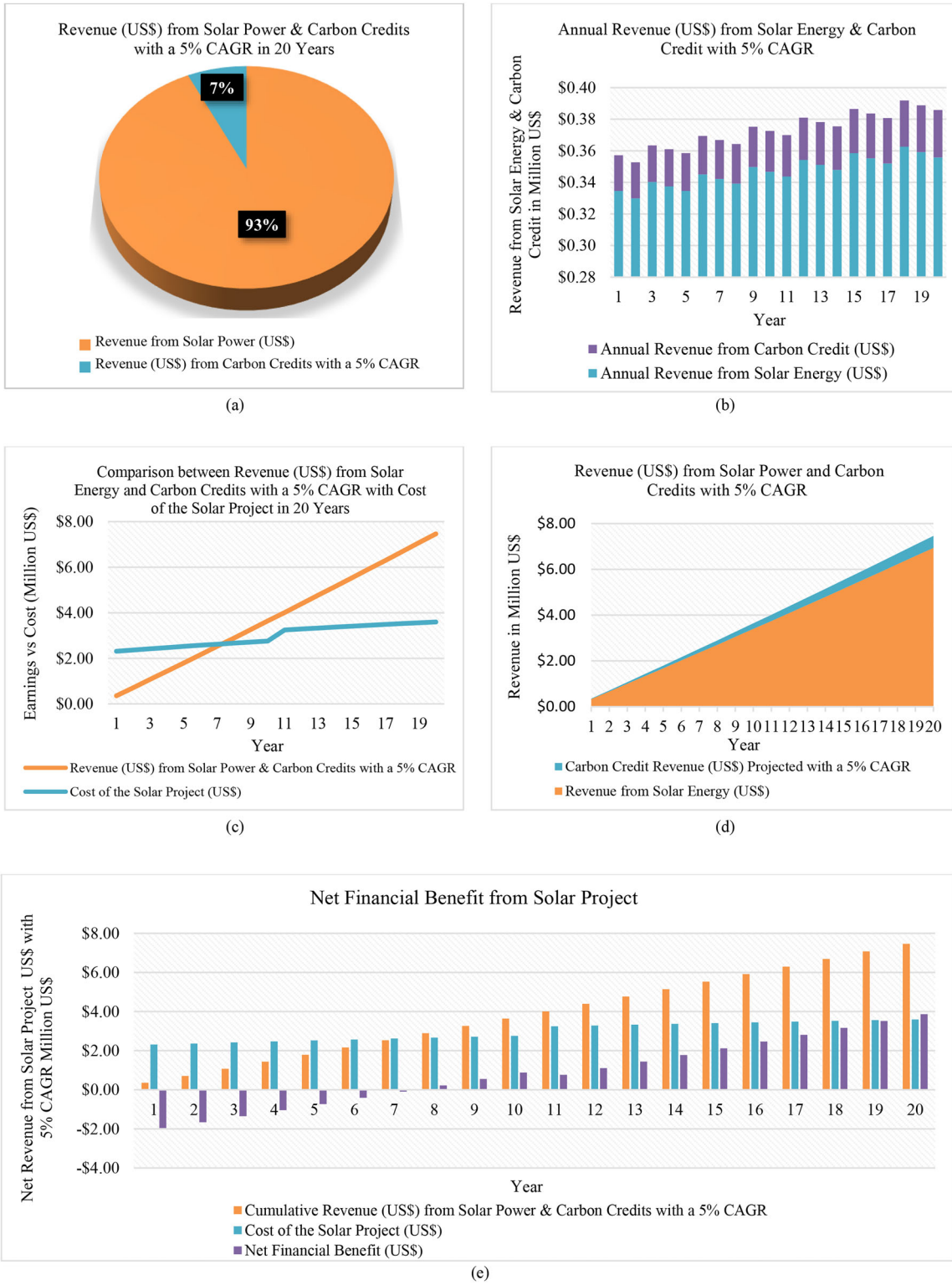


FIGURE 13 | cost-benefit analysis with a CAGR of 5% of carbon credit; (a) share of revenue (%) from solar energy and carbon credit; (b) annual revenue (US$) from solar energy and carbon credit; (c) revenue (US$) from (solar energy + carbon credits) Vs cost of the solar project in 20 years; (d) revenue (US$) from solar power and carbon credits in 20 years; and (e) net financial benefit (US$) with a CAGR of 5%.

the revenue from the carbon credit has been incorporated, it resulted in a reduced LCOE or tariff. Nevertheless, LCOE without considering carbon credit earnings would remain higher.

5.6 | Sensitivity Analysis

The study found that, at a CAGR of 5.0%, the NPV amounts to US$3.86 million, with an IRR of 9.85% and a short payback

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TABLE 9 | Cost-benefit analysis results for different compound annual growth rate (CAGR%) of carbon credit (in present value).

CAGR (carbon credit)	NPV (US$)	IIR (%)	Payback period (yrs)	LCOE
5.0%	US$3.86 million	9.85	8.35	US$0.0179/Kwh or BDT 2.18/Kwh
3.0%	US$3.77 million	9.61	8.45	US$0.0184/Kwh or BDT 2.25/Kwh
1.5%	US$3.72 million	9.46	8.55	US$0.0187/Kwh or BDT 2.29/Kwh
0.0%	US$3.67 million	9.31	8.52	US$0.019/Kwh or BDT 2.32/Kwh
(−) 10.0%	US$3.50 million	8.65	8.47	US$0.0200/Kwh or BDT 2.44/Kwh
(−) 25.0%	US$3.42 million	8.17	8.41	US$0.0204/Kwh or BDT 2.50/Kwh
(−) 50.0%	US$3.38 million	7.83	8.34	US$0.0207/Kwh or BDT 2.52/Kwh

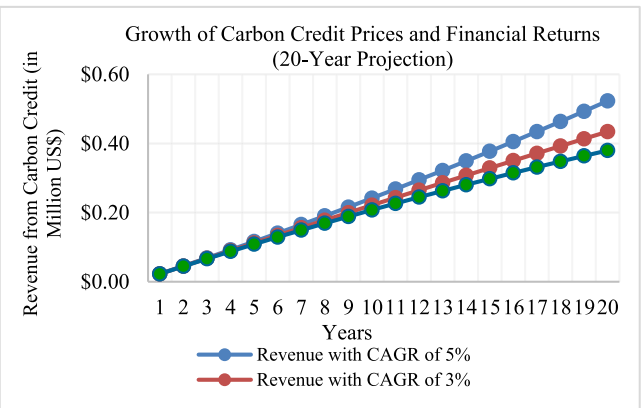


FIGURE 14 | Sensitivity analysis: financial returns (US $) (20-year projection) with the growth of carbon credit prices.

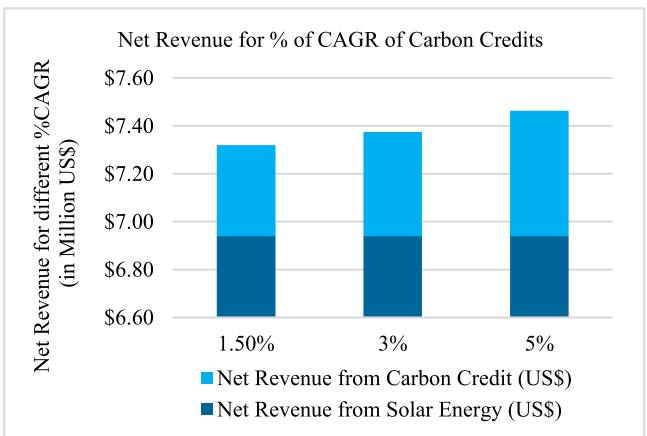


FIGURE 15 | Sensitivity analysis: comparison of revenues (US $) with 1.5%, 3% and 5% of CAGR of carbon credits.

period of 8.35 years. As the percentage of CAGR decreases, the NPV and IRR decline while the payback period increases. For CAGRs of 3.0% and 1.5%, the NPV stands at US$3.77 million and US$3.72 million, respectively, while the IRR is recorded at 9.61% and 9.46%. Moreover, the payback periods are 8.45 and 8.55 years,, respectively. Therefore, it is evident that as the percentage of CAGR of carbon credit declines, the NPV and IRR drop accordingly. However, the payback period becomes longer with a decreasing CAGR rate. The LCOE is calculated as US$0.0179/Kwh for a CAGR of 5.0%, US$0.0184/Kwh for a CAGR of 3.0% and US$0.0187/Kwh for a CAGR of 1.5%, which are significantly lower than the current tariff rate. Figure 14 below illustrates net revenue from carbon credits for assumed CAGRs of 1.5%, 3% and 5% over a 20-year projection period as part of the sensitivity analysis.

The Figure 15 below illustrates the amount of revenue generated in US$ from the solar NMS over a span of 20 years, while energy generation is held constant in all scenarios, the revenue from carbon credits varies with assumed CAGRs of 1.5%, 3%, and 5%.

The spider chart in Figures 16 and 17 shows the sensitivity of both NPV and IIR to the changes in the CAGR of carbon credits. The sensitivity analysis is performed based on a baseline carbon credit price of US$5 per unit. The results indicate that in favourable market conditions (i.e., during the growth of carbon credit prices) (+5% CAGR), the project yields an NPV of US$3.87 million and an IIR of 9.85%. Conversely, with a significant decline in carbon credit prices, the NPV and IIR remain positive at US$3.39 million and 7.83%, respectively. Despite the reduction of carbon credit prices over time, the project continues to generate positive returns. Therefore, it clearly indicates that the financial feasibility of the project does not critically depend on the fluctuations of carbon credit prices.

6 | Limitations

The study conducts the cost-benefit analysis assuming a rooftop solar installation capacity of 70% of the sanctioned load, in line

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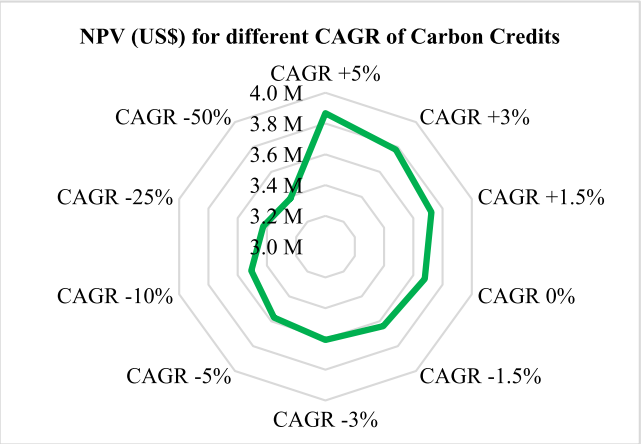


FIGURE 16 | Sensitivity analysis: impact on project NPV for different carbon credit price rate.

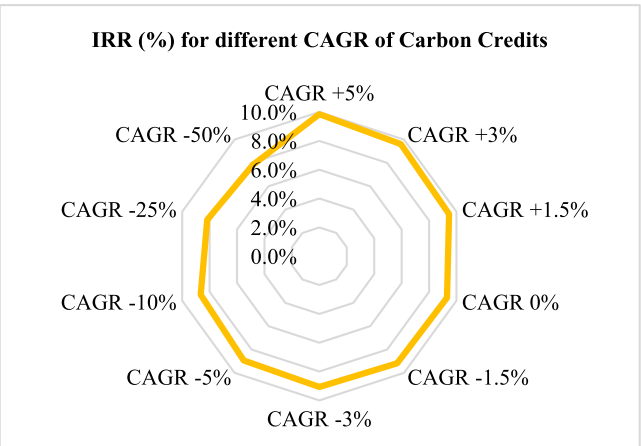


FIGURE 17 | Sensitivity analysis: impact on project IRR for different carbon credit price rate.

with the 2018 Net Metering Guideline. However, the 2025 guideline removes this cap, allowing up to 100% installation. Adopting the full 100% capacity could potentially enhance the project's economic performance, which could potentially improve the project's economic performance. Nevertheless, using 70% is a conservative, grid-friendly approach which reduces the risk of overgeneration and minimises stress on weak rural distribution networks. Moreover, the study combines 5,424 facilities into a single aggregated system; thus, the reported NPVs (in US$) reflect an average across all sites. Individual facility performance may differ due to variations in load profiles, rooftop availability, and local grid conditions.

7 | Recommendations

Based on the thorough review of recent projects [16, 17, 22, 24, 27, 41, 42, 65], the study figured out several gaps in the existing Net Metering Guideline-2018 of Bangladesh, where some of them have been addressed in the newly introduced Net Metering Guideline-2025. The latest guideline broadens eligibility by including three-phase consumers, removes the 10 MW capacity limitation and permits installation capacities up to 100% of the sanctioned load.

The new (2025) policy does not address key issues identified in the financial model between the consumer and the EPC, including limited control for both the consumer, billing and payment disputes, marketing restrictions on excess power sales, long-term PPA rigidity and weak asset recognition affecting grid integration. The following incentives can encourage greater adoption of net energy metering (NEM) systems and boost growth of a clean energy-driven economy:

- Introduction of incentives for prosumers, such as awarding bonus payment for grid-fed energy.
- Exemption of taxes and wheeling charges.
- Capacity building mechanism to develop skilled installers and auditors by providing necessary training and certifications.
- Providing prosumers with carbon credits trading facilities.

8 | Conclusion

In this study, a thorough cost-benefit analysis has been performed to determine several parameters like IRR, NPV, payback period and LCOE utilising carbon credits with a rooftop solar net metering system for the rural healthcare infrastructures of Bangladesh. The analysis was done using the net metering calculator developed by SREDA in accordance with the Net Metering Guideline-2025 of Bangladesh. Further analysis was conducted considering carbon credits as the SREDA calculator does not consider the carbon credit in its calculation.

The estimated cost of the project in its lifecycle was around US$3.59 million as in present value. In accordance with the results of the SREDA net metering calculator, the project yields around 171.48 (GWhr) of energy and reduces around 82,999.02 MT of carbon during a 20 year period. The amount of revenue generated from solar energy (Kwh) was estimated around US$6.94 million in present value considering the BREB's current bulk tariff rate. However, as SREDA's calculator does not consider the revenue from carbon credits, the revenue from the carbon credit sales has been integrated in this study. Since carbon credit market is fluctuating in nature, for an accurate projection of carbon credit revenue, several values of CAGR (1.5%, 3%, 5%, 0%, −10%, −25% and −50%) were considered based on future market trends. Adding carbon credit revenue with energy (Kwh), the project generates US$7.319 million in revenue for a CAGR of 1.5%, US$7.463 million for a CAGR of 3% and US$7.374 million for a CAGR of 5%, leading to a significant financial gain.

The sensitivity analysis indicates that with the increase of carbon credit price, the NPV and IRR also increase and payback period and LCOE decrease as expected and vice-versa. In each case the LCOE was reduced and ranged between US$0.0179/Kwh and US$0.0187/Kwh which is significantly lower compared to the current LCOE of 0.0358 US$/KWh as charged by the BREB. Even with a decrease in carbon credit prices by −10%, −25% and −50% CAGR, the LCOE remains comparatively lower. Therefore, it indicates that the project's financial viability is not significantly

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affected by fluctuations in carbon credit prices. The project demonstrates a positive NPV, robust IRR, low LCOE and a reasonable payback period even under declining carbon credit prices (e.g., −25% and −50% CAGR), providing strong evidence of its financial viability independent of carbon market fluctuations.

This places it as the most cost-effective source of electricity generation after hydro power in Bangladesh, ensuring a highly efficient energy generation system. Consuming grid electricity without solar NMS, the rural healthcare facilities would have to incur approximately US $9.9 million (in present value) over the next 20 years under the current tariff structure. The study also reviews the existing gaps in the latest Net Metering Guidelines-2025 of Bangladesh and proposes several recommendations to meet the existing gaps.

The prospect of trading carbon credits has also been studied, as based on current market trends and existing installed generation capacity of renewable energy, Bangladesh could generate 40,656 units of carbon credits annually, which will generate a significant amount of profit.

The key contributions of this study are: (1) the application of solar-based NMS modelling to the underexplored rural healthcare sector of Bangladesh; (2) the development of an innovative financial model integrating carbon credit revenues; (3) the review of the 2018 and 2025 Net Metering Guidelines to identify gaps and provide evidence-based, data-driven recommendations for strengthening future regulatory frameworks; (4) the development of a practice for embedding carbon credit components in renewable energy projects to maximise green energy benefits; (5) the provision of a strategic roadmap toward a clean, sustainable, and low-carbon energy transition in Bangladesh and (6) the dissemination of evidence-based insights to all concerned stakeholders, including utilities, customers and investors, with evidence-based insights for scaling renewable NMS initiatives effectively.

This rooftop solar NMS project is efficient in utilising spaces, requires minimal maintenance, environment friendly and cost-effective in the long run. Such projects can play a significant role in promoting energy security through reducing reliance on fossil fuels and help to meet carbon footprint reduction goals. It is high time for Bangladesh to develop a policy or framework that obligates the heavy industries to purchase electricity from renewable energy. Provision could be kept under the policy or framework that the government owned utilities, agencies and captive power producers responsible for carbon emissions shall be mandated to purchase a certain amount (percentage) of electricity from renewable energy annually.

One of the limitations of this solar powered grid-tied NMS system is it does not include battery storage; thus, it would not be able to operate during the grid outages. However, this system to some extent improves reliability as it reduces daytime grid dependency. Therefore, future studies could be performed to examine the techno-economic impact of a hybrid solar NMS system with battery storage to enhance reliability and address the grid outages effectively.

In the future, a small-scale pilot NMS project with phased investment could be explored using stochastic optimisation frameworks to manage real-world uncertainties [78]. Moreover, further studies could focus on developing an optimised network-level distributed system to manage flexible energy loads across multiple healthcare facilities, enabling better synchronisation with solar power generation [79].

Author Contributions

Syed Enam Reza: conceptualisation, data curation, formal analysis, investigation, methodology, resources, validation, visualisation, writing – original draft, writing – review and editing. **H. M. Imran Hassan:** conceptualisation, formal analysis, supervision, writing – review and editing. **Khondaker Abdullah Al Mamun:** conceptualisation, formal analysis, funding acquisition, project administration, resources, supervision, visualisation, writing – review and editing. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy or ethical restrictions.

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